

Weekly Electricity Consumption: Time- and Frequency-Domain Analysis of T200

MATH5845 Time Series Group Project: Electricity Group
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1. Introduction

This project analyses the T200 weekly electricity-consumption series from the Monash Time Series Forecasting Repository. The series contains 156 weekly observations from 1 January 2012 to 21 December 2014. Our aim is to identify the main dependence and seasonal features in T200 and select a parsimonious model that captures them well. We compare ARMA, ARIMA and SARIMA candidates using the time plot, ACF/PACF, information criteria, estimated coefficients and residual diagnostics. A frequency-domain analysis is then used as an independent check of the annual periodic behaviour suggested by the time-domain analysis.

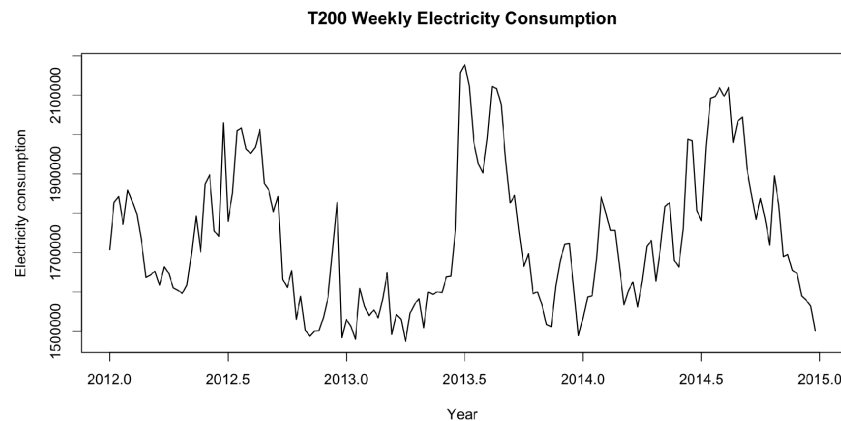
2. Data Preparation and Exploratory Analysis

The original dataset contains multiple weekly electricity-consumption series, from which T200 was extracted for univariate analysis. T200 contains 156 observations with a natural annual period of approximately 52 weeks and has no missing values. No observations were removed or adjusted as outliers during preprocessing; potentially unusual observations were instead assessed later through the fitted-model residual diagnostics.

Figure 1 shows substantial week-to-week variation together with a recurring pattern across the three years. Consumption tends to rise around the middle of each year before declining later in the year, suggesting an annual seasonal component. There is no clear persistent upward or downward trend. The mean weekly consumption is approximately 1.74 million, with a standard deviation of approximately 0.18 million and values ranging from approximately 1.47 to 2.18 million. The magnitude of the fluctuations is not completely uniform over time: variability appears somewhat greater around the beginning of 2013 than during much of 2012. However, there is no clear systematic relationship in which variability consistently increases over time. We therefore investigated a Box-Cox transformation as a possible variance-stabilising transformation.

The sample ACF decays slowly and remains substantial across many lags, including seasonal lags, which confirms strong serial dependence. The PACF has a dominant spike at the first lag followed by much smaller values. Together, the ACF and PACF suggest strong short-run dependence and annual seasonal structure, but they do not identify a unique ARMA order. We therefore use them as guides when choosing candidate models rather than applying a mechanical cutoff rule.

(a) Original weekly series



(b) Original-series ACF and PACF

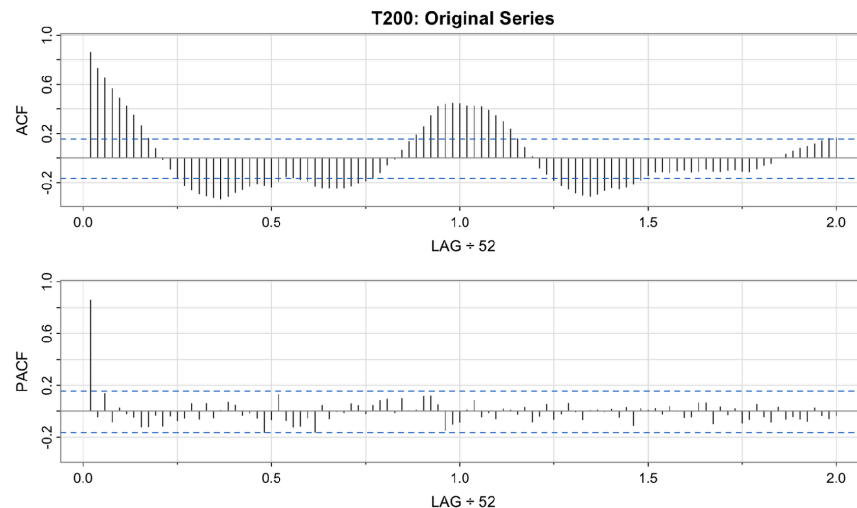


Figure 1. Original T200 series and its sample ACF/PACF.

2.1 Differencing and Stationarity

Differencing was assessed mainly from the time plot and ACF/PACF. As supplementary numerical checks, $\text{ndiffs}(y)$ suggested $d = 0$ and $\text{nsdiffs}(y)$ suggested one seasonal difference, $D = 1$. The ADF and KPSS tests gave mixed evidence for the original and seasonally differenced series, so we did not use either test mechanically to choose the differencing order. Both $\{d = 0, D = 1\}$ and $\{d = 1, D = 1\}$ were therefore considered during model fitting. ADF and KPSS address stationarity in the level of the series; they do not test whether its variance changes over time. We considered possible changes in variance separately when assessing Box-Cox transformations.

Series	ADF p	KPSS p	Interpretation
Original	0.3168	>0.10	Mixed evidence
First difference	<0.01	>0.10	Supports stationarity
Seasonal difference	0.0301	<0.01	Mixed evidence
First + seasonal difference	<0.01	>0.10	Supports stationarity

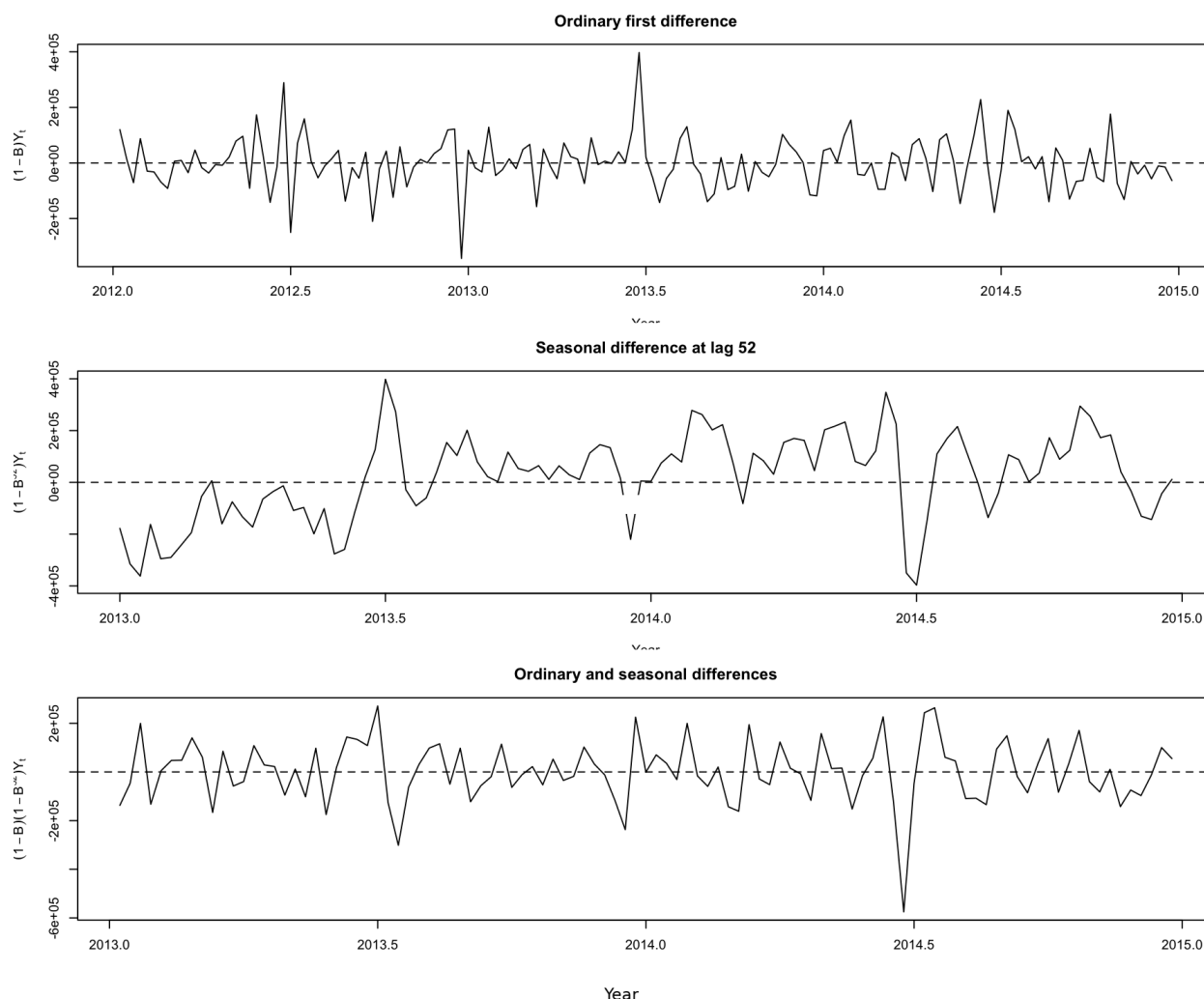


Figure 2. Ordinary, seasonal and combined differences considered during model identification.

2.2 Box-Cox Transformation: Joint Likelihood Approach and Guerrero Criterion

Initially, we considered the conventional likelihood-based approach to selecting the Box-Cox parameter. Under this approach, each candidate value of λ transforms the series, a time-series model is fitted to the transformed data, and λ is chosen to maximise the Gaussian likelihood. Because the transformation can affect the fitted dependence structure, we then extended this idea by estimating λ jointly with the SARIMA parameters rather than treating the transformation as separate from the model. Importantly, when likelihoods from different Box-Cox transformations are compared on the original-data scale, the change-of-variables Jacobian must also be included. Without this Jacobian adjustment, the likelihoods (and as a result the AIC, AICc and BIC values) are not directly comparable across different values of λ because the transformation itself changes the scale and density of the observations.

For the leading SARIMA structures, the joint-likelihood approach gave $\lambda \approx -0.38$. However, the fitted residuals were far from Gaussian. For the Shapiro-Wilk test, the null hypothesis is that the residuals are normally distributed; here W was approximately 0.14-0.21 with p -values below 10^{-20} , so this null hypothesis was strongly rejected. Because the joint method selects λ using a Gaussian likelihood, this assumption was poorly supported by the fitted models. We therefore treated the joint-likelihood result as a sensitivity check rather than as the main transformation choice.

Since the main purpose of the Box-Cox transformation here was variance stabilisation, Guerrero's criterion was then used as the main transformation rule. It gave $\lambda = 1.999924$, essentially identical to the lecturer's value of 1.999927. We therefore carried $\lambda \approx 2$ forward and judged it by the fitted-model diagnostics and information criteria.

All models and diagnostics reported below are generated from the same R codebase and the submitted T200 dataset.

3. Time-Domain Model Identification and Fitting

Rather than relying on a single automatic model, we fitted a broad set of models from the ARMA family. These included low-order ARMA models on the original series, nonseasonal ARIMA models, and SARIMA models with annual period 52. The ARMA and ARIMA fits were useful benchmarks, while the seasonal SARIMA models were examined more closely because the exploratory analysis showed clear annual structure. For the SARIMA candidates we considered both $\{d = 0, D = 1\}$ and $\{d = 1, D = 1\}$, reflecting the mixed differencing evidence in Section 2.1.

As a supplementary benchmark, `auto.arima()` selected SARIMA(0,1,3)(1,1,0)[52]. The candidate models below were nevertheless identified and compared independently rather than restricting the analysis to the automatically selected specification.

Among the SARIMA candidates, two untransformed models were especially competitive: SARIMA(0,1,3)(1,1,0)[52] and SARIMA(2,1,1)(1,1,0)[52]. They had the best combination of AICc/BIC values and residual autocorrelation diagnostics among the untransformed SARIMA fits. Other candidates were not taken forward to the detailed comparison because they had clearly weaker information criteria, less convincing coefficient estimates, or poorer residual behaviour.

These same two models were then refitted using the Guerrero Box-Cox transformation so that the effect of the transformation could be compared directly.

Raw SARIMA	AICc	BIC	Ljung-Box p (20)	Shapiro p	Coefficient evidence
(0,1,3)(1,1,0)[52]	2682.16	2694.71	0.958	0.00243	MA(1) not sig.; MA(3) borderline
(2,1,1)(1,1,0)[52]	2682.89	2695.44	0.931	0.000874	All AR/MA coefficients sig.

4. Frequency-Domain Analysis

We also examined the series in the frequency domain to see whether the periodic behaviour suggested by the time-domain analysis appeared there as well. While the ACF describes dependence across time lags, the periodogram shows how variance is distributed across frequencies, making recurring cycles easier to identify. Since T200 contains only 156 weekly observations, or about three annual cycles, the estimated peak locations should be interpreted approximately rather than exactly due to lower confidence.

The largest distinct peak in the raw periodogram occurs at about 0.975 cycles per year, which corresponds to roughly 53.3 weeks. A second clear peak appears at about 1.95 cycles per year, or 26.7 weeks. Because 1.95 is almost exactly twice 0.975, this second peak is best interpreted as the second harmonic of the annual cycle rather than as a separate 27-week seasonal pattern. Therefore, the main peak is consistent with an approximately yearly cycle and evidences the seasonal period $s = 52$ choice in the SARIMA section.

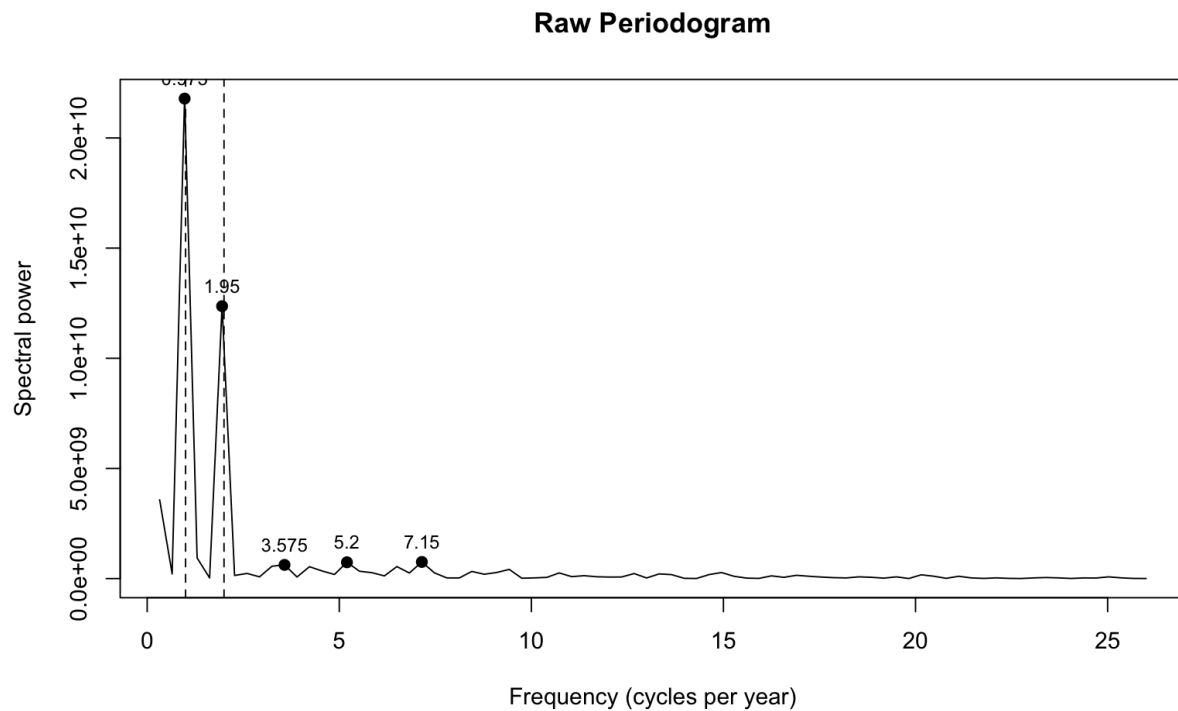


Figure 3. Raw periodogram. The dominant local peak occurs near one cycle per year, with a second harmonic near two cycles per year.

In order to check statistical robustness, we removed the fitted linear trend and recalculated the periodogram. The two main peaks stayed at very similar frequencies, suggesting that the annual pattern and its harmonic are not being created by the weak linear trend.

Because the raw periodogram is quite noisy, we also applied Daniell smoothing. The advantage of this is that it reduces the variance of the spectral estimate and makes the broader pattern easier to see. After smoothing, most of the spectral power is still concentrated in the low-frequency region at around one cycle per year. The sharp peaks at 0.975 and 1.95 cycles per year are less distinctive, but this is expected from smoothing.

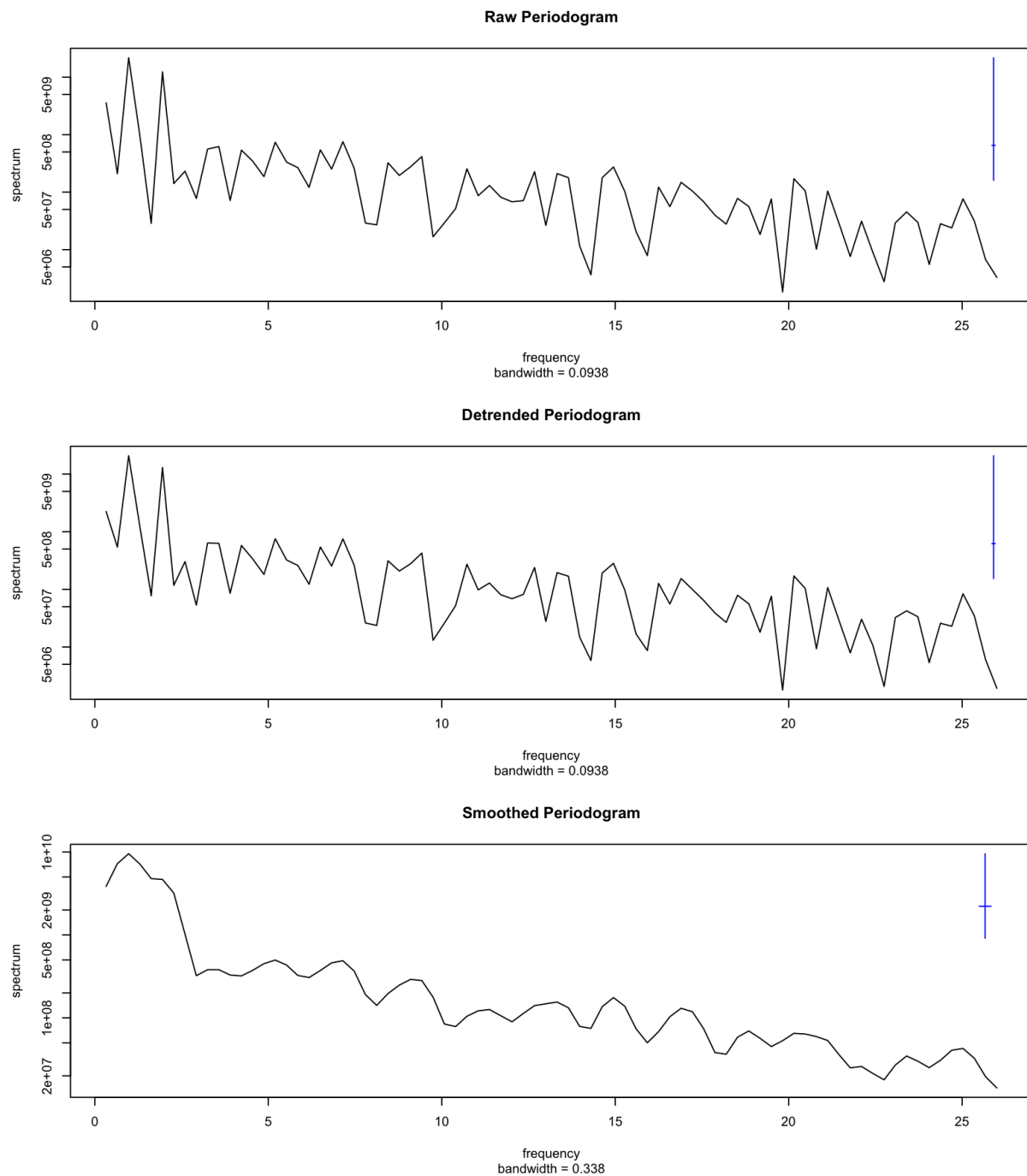


Figure 4. Raw, detrended and Daniell-smoothed periodograms. Spectral power remains concentrated around the approximately annual region after detrending and smoothing.

For comparison, we also fitted a parametric $AR(p)$ spectrum to the detrended series using the Yule-Walker method, with p chosen by AIC. This selected an $AR(1)$ model. Its spectral density captures the broad concentration of power at lower frequencies, but it does not reproduce the more concentrated power around the approximately annual frequency seen in the non-parametric estimates. In this dataset, the $AR(1)$ spectrum is useful for describing the broad low-frequency dependence, while the periodogram gives a clearer picture of the annual component.

Smoothed Periodogram vs AR(1) Spectrum

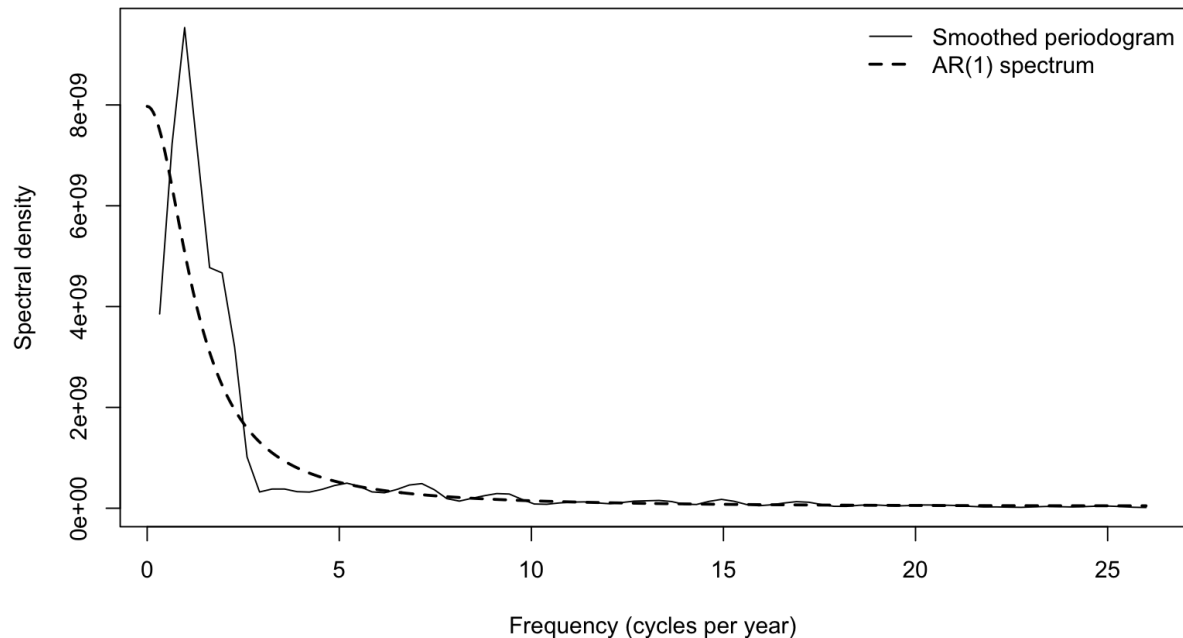


Figure 5. Smoothed periodogram compared with the fitted AR(1) parametric spectral estimate.

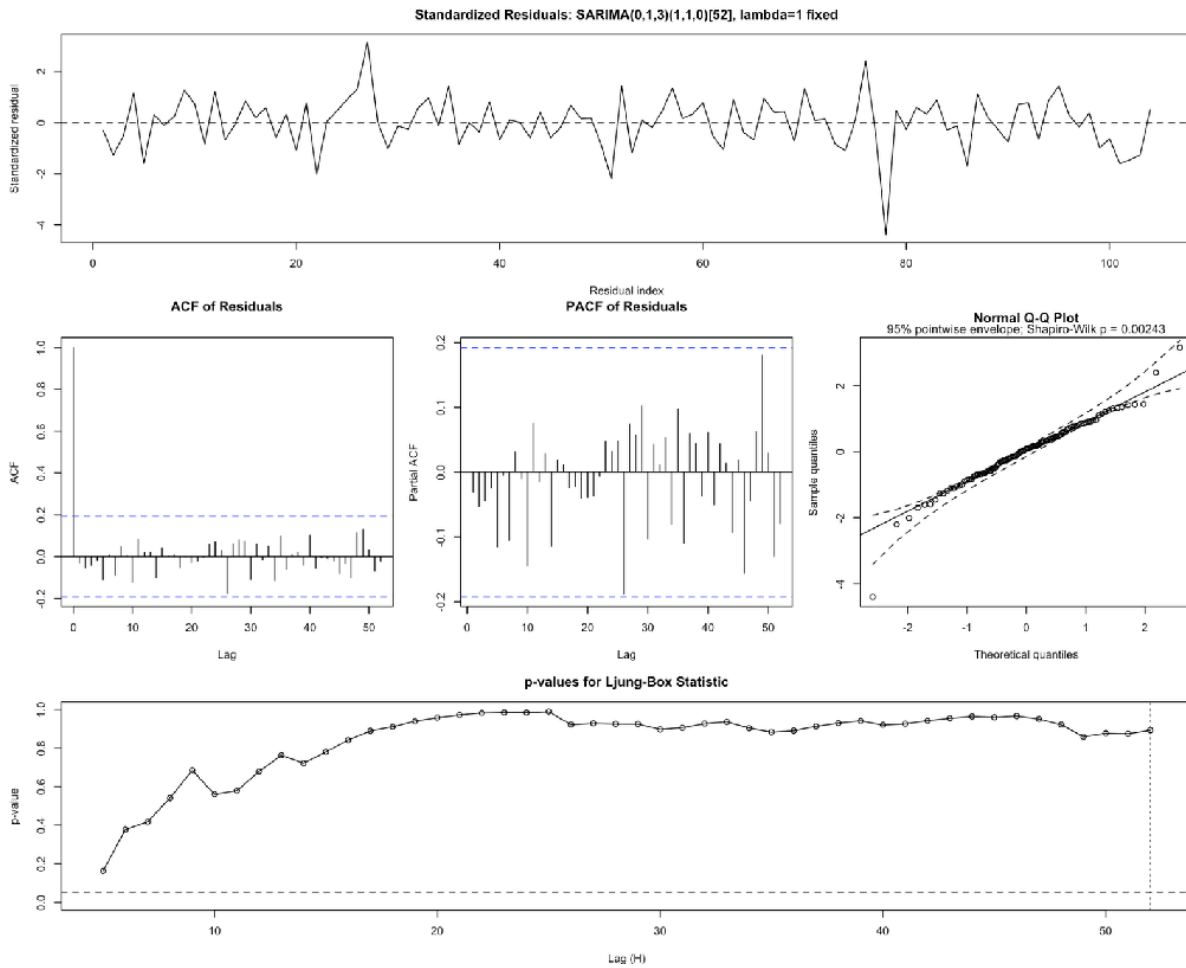
Taken together, the frequency-domain results tell the same basic story as the time-domain section. There is a dominant component near one cycle per year, consistent with the seasonal period $s = 52$ used in the SARIMA models, and the peak near two cycles per year is consistent with a second harmonic. This also remains after detrending and smoothing, although with only three years of data we do not place too much weight on the exact positions of the peaks.

5. Residual Diagnostics and Top-Model Comparison

For models with seasonal differencing, we removed the first 52 residuals before the diagnostic checks, following the lecturer's Form 2 feedback. Those initial residuals are affected by the start of the lag-52 differencing and are not directly comparable with the later residuals. We then checked the remaining residuals using the residual plots, ACF/PACF, the Ljung-Box test, and the Q-Q and Shapiro-Wilk normality checks. We report the Ljung-Box test at lag 20: after differencing there are about 103 usable observations, so 20 is roughly one-fifth of the remaining sample and is close to the default lag used by `checkresiduals()` for a series of this length. The diagnostic panels also show Ljung-Box p-values over a range of lags, so the conclusion is not based on one lag alone. The main requirement is that the residuals show no meaningful remaining autocorrelation.

5.1 Top models on the original scale

(a) SARIMA(0,1,3)(1,1,0)[52]



(b) SARIMA(2,1,1)(1,1,0)[52]

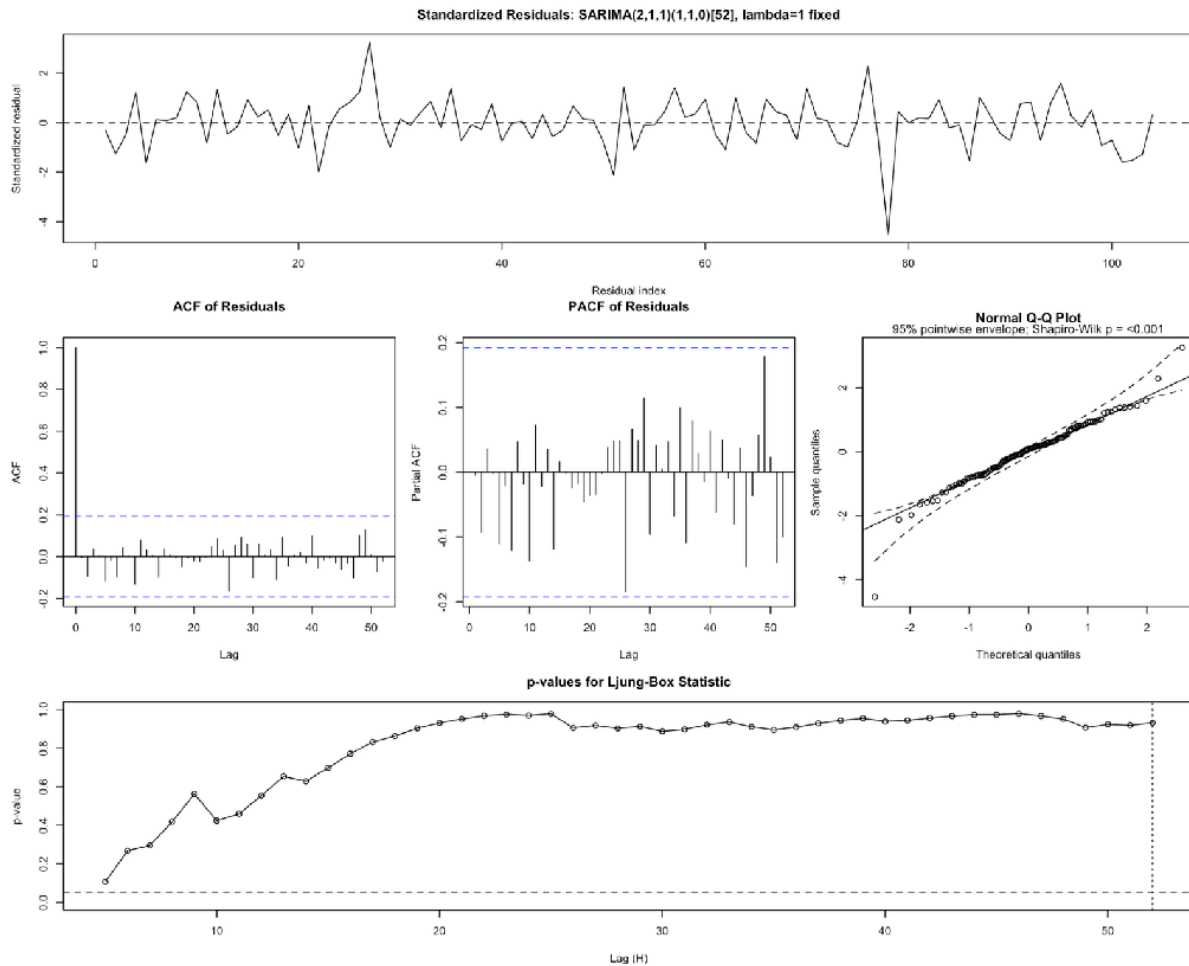


Figure 6. Residual diagnostics for the two leading untransformed SARIMA models.

Both untransformed top models perform well on the residual autocorrelation checks. Their residual ACFs contain little remaining structure, and the Ljung-Box p-values at lag 20 are 0.958 for SARIMA(0,1,3)(1,1,0)[52] and 0.931 for SARIMA(2,1,1)(1,1,0)[52]. Both values are well above 0.05, so there is no evidence of important remaining autocorrelation. The Shapiro-Wilk tests do reject normality ($p = 0.00243$ and $p = 0.000874$), and the Q-Q plots suggest that this is mainly due to departures in the tails and one large negative residual.

The two models are therefore difficult to separate using residual autocorrelation alone. SARIMA(0,1,3)(1,1,0)[52] has the slightly smaller AICc, but its MA(1) coefficient is not significant and MA(3) is only just significant at the 5% level. In SARIMA(2,1,1)(1,1,0)[52], all fitted AR and MA coefficients are significant. Meanwhile, both have a very similar overall fit and residual behaviour.

5.2 Joint-Likelihood Box-Cox

Before using Guerrero's method, we also tried estimating λ jointly with the SARIMA parameters. The corrected estimates were around -0.38. The Ljung-Box checks showed little remaining autocorrelation, but the Shapiro-Wilk tests strongly rejected Gaussian residuals ($W \approx 0.14$ -0.21, $p < 10^{-20}$). Because this approach chooses λ by maximising a Gaussian likelihood, the normality failure makes the lower likelihood-based criteria less persuasive. We therefore kept this as supportive evidence rather than using it to choose the transformation.

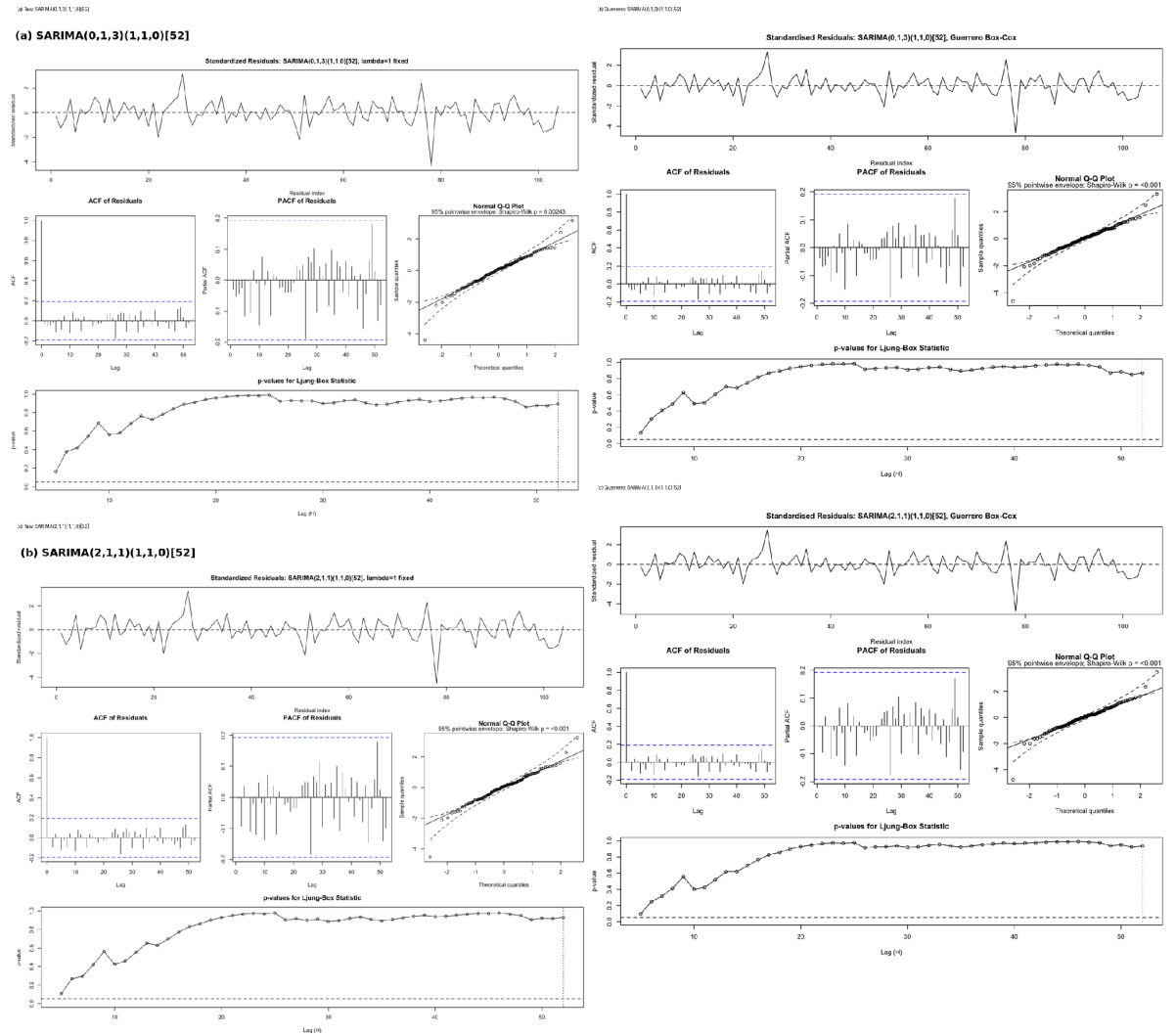
Initial joint-likelihood fit	λ	Joint AICc	Joint BIC	Shapiro p	Coefficient evidence
BC (0,1,3)(1,1,0)[52]	-0.3826	2676.6	2691.6	7.1e-22	MA(1), MA(3) not sig.
BC (2,1,1)(1,1,0)[52]	-0.3811	2677.4	2692.3	7.6e-22	All AR/MA coefficients sig.
BC (2,1,1)(0,1,0)[52]	-0.3439	2687.9	2700.4	4.2e-21	All AR/MA coefficients sig.

The lower AICc values from this joint-likelihood check were therefore not treated as sufficient reason to use $\lambda \approx -0.38$. Instead, we moved to Guerrero's criterion, which is aimed directly at stabilising variance.

5.3 Guerrero Box-Cox Comparison

Guerrero's method gave $\lambda = 1.999924$, so we refitted the two leading SARIMA structures using $\lambda \approx 2$. We also kept SARIMA(2,1,1)(0,1,0)[52] as an additional transformed comparison because it had been considered earlier in the analysis. The transformed models still showed little residual autocorrelation, but this was not an improvement over the untransformed fits. Normality also remained rejected, and the Q-Q plots continued to show tail departures. After including the Box-Cox Jacobian so that likelihood-based criteria are comparable across transformations, the transformed versions also had larger AICc and BIC values than their untransformed counterparts. For example, the AICc for SARIMA(0,1,3)(1,1,0)[52] increased from 2682.16 to 2693.40, while SARIMA(2,1,1)(1,1,0)[52] increased from 2682.89 to 2693.87. We therefore did not retain the Guerrero transformation for the final model.

Model	Transformation	AICc	BIC	Shapiro p	Coefficient evidence
(0,1,3)(1,1,0)[52]	Raw / $\lambda=1$	2682.16	2694.71	0.00243	MA(1) not sig.; MA(3) borderline
(2,1,1)(1,1,0)[52]	Raw / $\lambda=1$	2682.89	2695.44	0.000874	All AR/MA coefficients sig.
(0,1,3)(1,1,0)[52]	Guerrero $\lambda \approx 2$	2693.40	2708.19	0.000305	MA(1) not sig.
(2,1,1)(1,1,0)[52]	Guerrero $\lambda \approx 2$	2693.87	2708.66	0.000098	All AR/MA coefficients sig.
(2,1,1)(0,1,0)[52]	Guerrero $\lambda \approx 2$	2701.46	2713.99	0.000061	All AR/MA coefficients sig.



6. Model Comparison and Final Model Selection

The two leading untransformed models are very close on the information criteria. SARIMA(0,1,3)(1,1,0)[52] has AICc 2682.16 and BIC 2694.71, while SARIMA(2,1,1)(1,1,0)[52] has AICc 2682.89 and BIC 2695.44. The AICc difference is only $2682.89 - 2682.16 = 0.73$, which is small, so the information criteria do not provide a strong reason to choose one model over the other.

The coefficient estimates provide a clearer difference. In SARIMA(0,1,3)(1,1,0)[52], MA(1) is not significant at the 5% level ($p = 0.183$), while MA(3) is only just significant ($p = 0.0489$). In SARIMA(2,1,1)(1,1,0)[52], both AR coefficients, MA(1), and the seasonal AR(1) coefficient are significant at 5%.

The residual checks are similarly strong for both models. At lag 20, the Ljung-Box p-values are 0.958 and 0.931, so neither test gives evidence of remaining autocorrelation. The residual ACFs agree with this conclusion. Both models fail the Shapiro-Wilk normality test, but the Q-Q plots suggest that the main departures occur in the tails. Because normality is not strictly required to conclude a good model fit, it does not clearly favour one model over the other.

The Guerrero transformation also did not improve the comparison. In particular, the transformed models had larger Jacobian-adjusted AICc/BIC values and still failed the normality checks. For the preferred SARIMA(2,1,1)(1,1,0)[52] model, the fitted AR roots are outside the unit circle (with smallest root of 1.016). Thus, after the required differencing, the fitted AR part satisfies the course's causality/stationarity condition, although the root close to one indicates longer-term persistence..

Overall, we select SARIMA(2,1,1)(1,1,0)[52] on the original scale. Its AICc is only 0.73 higher than the minimum AICc among the compared models, all fitted AR and MA coefficients are significant, and the residuals show no meaningful remaining autocorrelation. Our choice therefore reflects the combined evidence from information criteria, coefficient estimates and residual diagnostics rather than simply choosing the model with the smallest AICc.

7. Discussion and Limitations

The main limitation is the short sample: 156 weekly observations contain only about three annual cycles. This means the annual seasonal pattern is estimated from only three repetitions, and the frequency-domain analysis has limited resolution at low frequencies. It also makes it harder to distinguish confidently between closely competing SARIMA models. The residual Q-Q plots and Shapiro-Wilk tests show departures from normality, especially in the tails, so likelihood-based comparisons that rely on Gaussian innovations should be interpreted cautiously. Variance also appears to change somewhat across the sample, although there is no simple systematic relationship between the level and variance. Finally, the selected model is assessed on this single series and sample period, without a separate out-of-sample forecasting evaluation, so the residual diagnostics do not by themselves establish how well the model will forecast future observations.

8. Conclusion

We compared a broad set of ARMA, ARIMA and SARIMA models for the T200 weekly electricity series. The strongest untransformed models were SARIMA(0,1,3)(1,1,0)[52] and SARIMA(2,1,1)(1,1,0)[52], while Box-Cox transformations did not improve the overall model comparison. We select SARIMA(2,1,1)(1,1,0)[52] because it is essentially tied on AICc with the best competing model, all of its fitted AR and MA coefficients are significant, and its residuals show little remaining autocorrelation. The frequency-domain analysis independently supports the same annual seasonal structure through a dominant component near one cycle per year.